MIL-P-82650(OS)
20 October 1976
Superseding
WS 14776
(See Section 6)

MILITARY SPECIFICATION

PLASTIC MOLDING MATERIAL, GLASS PHENOLIC

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of glass phenolic plastic molding material.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposals, form a part of the specification to the extent specified herein.

SPECIFICATIONS

Federal

RR-S-366	Sieve, Test

Military

MIL-P-116	Preservation-Packaging, Methods of

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Ordnance Station, Standardization Division (611), Indian Head, MD 20640, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 9330

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:19Z
Check the source to verify that this is the current version before use.

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on the date of the invitation for bids or the request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 638-72	Tensile Properties of Plastics
ASTM D 695-69	Compressive Properties of Rigid Plastics
ASTM D 792-66	Specific Gravity and Density of Plastics by Displacement

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT
National Motor Freight Classification

(Application for copies should be addressed to American Trucking Associations, Attn: Tariff Order Section, 1616 P Street, Washington, DC 20036.)

UNIFORM CLASSIFICATION COMMITTEE, AGENT
Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Preproduction sample. Unless otherwise specified in the contract or order (see 6.2), a preproduction sample of the molding material is required and shall be manufactured using the methods and procedures proposed for production (see 4.4.1). The sample shall be tested as specified in 4.5 for the purpose of determining, prior to starting production, that the contractor's production methods are capable of producing molding material that complies with the technical requirements of the contract. No raw material process changes shall be made, subsequent to approval of the preproduction sample, without prior written approval of the procuring activity.

3.2 Material. The product shall be a glass phenolic plastic molding material, manufactured from Type E glass roving impregnated with a one-stage phenol formaldehyde resin.

Source: https://assist.dla.mil -- Downloaded: 202610-23T07:19Z
Check the source to verify that this is the current version before use.

3.3 Form. The product shall be furnished in a chopped form. One hundred percent of the material shall pass through a 1.0 inch (25.4 mm) sieve conforming to RR-S-366.

3.4 Storage life. The molding material shall be capable of meeting all requirements of this specification for a period of 6 months from the date of manufacture when stored at a maximum temperature of 4.5°C (40°F).

3.5 Chemical and physical properties.

3.5.1 Uncured molding material. The properties of the uncured molding material shall be in accordance with TABLE I.

3.5.2 Cured molding material. The properties of the cured molding material (see 4.5.1) shall be in accordance with TABLE II.

TABLE I. Uncured molding material properties.

Property	Values	
	minimum	maximum
Resin content, percent	35	41
Volatile content, percent	3	5

TABLE II. Cured molding material properties.

Property	Values	
	minimum	maximum
Specific gravity	1.75	-
Tensile strength, psi	4,000	-
Compressive strength, psi	10,000	-

3.6 Workmanship. The molding material shall be uniform in quality and shall be free from impurities and other defects that could adversely affect its use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:19Z
Check the source to verify that this is the current version before use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. Preproduction inspection (4.5)
b. Quality conformance inspection (see 4.6)

4.3 Inspection conditions. Unless otherwise specified (see 6.2), all inspections shall be performed under the following conditions:

a. Temperature: Room ambient 18 to 35°C (65 to 95°F)
b. Altitude: Normal ground
c. Vibration: None
d. Humidity: Room ambient to 95 percent relative, maximum

4.4 Sampling.

4.4.1 Preproduction sample. Unless otherwise specified in the contract or order (see 6.2), a preproduction sample of 50 kilograms (kg) minimum of molding material shall be manufactured in accordance with 3.1, packaged in accordance with Section 5 (2 unit packages, minimum), and forwarded to an activity designated by the procuring activity (see 6.2) for preproduction inspection as detailed in 4.5. Further production of the molding material by the contractor, prior to approval of the preproduction sample, shall be at the contractor's risk.

4.4.2 Lot. Unless otherwise specified (see 6.2), a lot shall consist of all molding material manufactured by the same process, from the same raw materials, by one manufacturer, at one plant, in a single production run, under essentially identical conditions, and offered for acceptance at one time. No lot of molding material shall contain more than one batch of any raw material.

4.4.3 Quality conformance inspection sampling. Quality conformance inspection samples shall be selected in accordance with inspection level S-4 of MIL-STD-105 from molding material prepared for delivery in accordance with 3.1. The sample unit shall be one unit package or container. Each sample shall consist of a minimum of 25 grams(g) of molding material obtained from filled sample containers, prior to sealing.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:19Z
Check the source to verify that this is the current version before use.

4.5 Preproduction inspection. One unit package of the preproduction sample, having been submitted to and having passed the quality conformance inspection of 4.6, shall be used to prepare the molded test panels of 4.5.1. The second unit package shall be stored, unopened, for 6 months at 4.5°C (40°F). The preproduction inspection shall consist of the tests of 4.5.2 through 4.5.5. Failure to meet any requirement of this specification shall be cause for rejection of the preproduction sample.

4.5.1 Preparation of molded test panels. Preproduction sample molding material shall be compression molded at 2000 + 200, - 0 psig for 30 to 60 minutes at 149 + 11 - 0°C (300 + 20, -0°F) with a post cure of 6 + 1, - 0 hours at 192 ± 14°C (325 ± 25°F) in an air circulating oven. Panels of sufficient size and number to perform the tests of 4.5.2, 4.5.3 and 4.5.4 shall be prepared.

4.5.2 Specific gravity. The specific gravity of the cured molding compound shall be determined in accordance with method A-1 of ASTM D 792-66. One test specimen from each test panel of 4.5.1 shall be prepared and tested.

4.5.3 Tensile strength. The tensile strength of the cured molding compound shall be determined in accordance with ASTM D 638-72, specimen Type I and grip separation speed B.

4.5.4 Compressive strength. The compressive strength of the cured molding compound shall be determined in accordance with ASTM D 695-69, using standard test specimens.

4.5.5 Storage life. After 6 months storage per 4.5, the second unit package of the preproduction sample shall be submitted to the quality conformance tests of 4.6 and the preproduction tests of 4.5.1 through 4.5.4. Failure to meet any requirement shall indicate failure of the storage life test and shall be cause for rejection of the preproduction sample.

4.6 Quality conformance inspection. Each sample obtained in accordance with 4.4.3 shall be subjected to the following tests. Failure of any sample to meet any requirement of this specification shall be cause for rejection of the lot. When specified in the contract or order (see 6.2.2), the contractor shall furnish test reports showing quantitative results for all quality conformance tests required by this specification for each lot of material.

4.6.1 Volatile content. The volatile content of the uncured molding material shall be determined as follows:

a. Weigh 5 to 6 grams(g) of sample to the nearest 0.001g into a tared crucible that has been previously ignited at 649 ± 14°C (1200 ± 25°F).

b. Heat the crucible and specimen in a circulating air oven at 192 ± 3°C (325 ± 5°F) for a minimum of 20 minutes.

c. Remove, cool in a desiccator and weigh (Do not remove specimen from crucible (see 4.6.2.a.)).

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:19Z
Check the source to verify that this is the current version before use.

d. Calculate the volatile content as follows:

$$\text{Volatile content (percent)} = \frac{W_1 - W_2}{W_1} \times 100$$

where: W_1 = weight of specimen before test (g)

W_2 = weight of specimen after test (g)

e. Repeat the procedure for a total of three 5g specimens from each sample.

f. Report each individual value and the average of the 3 tests.

4.6.2 Resin content. The resin content of the uncured molding material shall be determined as follows:

a. Place the crucible with the ash remaining from 4.6.1.c. directly into a muffle furnace.
b. Ignite at 649 ± 14°C (1200 ± 25°F) for 3 hours
c. Remove the crucible from the furnace, cool in a desiccator and reweigh.
d. Calculate the resin content as follows:

$$\text{Resin content (percent)} = \frac{W_2 - W_3}{W_1} \times 100$$

where W_1 = weight of specimen before volatiles test (g)

W_2 = weight of specimen after volatiles tests (g)

W_3 = weight of ash after resin test (g)

e. Repeat the procedure for each of the three specimens of 4.6.1.
f. Report each individual value and the average of the 3 tests.

5. PACKAGING

5.1 Preservation, packaging and packing. Unless otherwise specified in the contract or order (see 6.2), minimum requirements for preservation, packaging and packing shall be in accordance with MIL-P-116, method III. Container size shall be as specified in the contract or order (see 6.2). Containers shall comply with the Uniform Freight Classification Rules, National Motor Freight Classification Rules, or regulations of other carriers as applicable to the mode of transportation.

5.2 Marking. In addition to any special marking required by the contract or order (see 6.2), each container shall be marked in accordance with MIL-STD-129. Marking shall include, but not be limited to, the following information:

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:19Z
Check the source to verify that this is the current version before use.

a. Manufacturer's name and location.
b. Material trade name.
c. Net weight and volume.
d. Lot number, batch number and date of manufacture.
e. Shelf life and storage limitations (see 3.4).
f. Number and date of this specification.
g. Contract or purchase order number.

6. NOTES AND CONCLUDING MATERIAL

6.1 Intended use. Plastic molding material in accordance with this specification is intended for use in a molded component of a Terrier rocket motor.

6.2 Ordering data. Procurement documents should specify the following:

6.2.1 Procurement requirements.

a. Title, number and date of this specification.
b. Quantity required.
c. Inspection conditions when other than as specified (see 4.3).
d. When a preproduction sample is not required (see 3.1, 4.4.1 and 6.4).
e. Assigned activity for preproduction inspection (see 4.4.1).
f. Lot size if other than as specified (see 4.4.2)
g. Packaging requirements if other than as specified (see 5.1).
h. Size of container required (see 5.1).
i. Any special markings required (see 5.2).

6.2.2 Contract data requirements. Items of deliverable data required by this specification are cited in the following paragraph herein:

Paragraph	Data Requirement	Applicable DID*
4.6	Quality conformance inspection data	-

*DIDS (Data Item Descriptions/DD Form 1664) for the above requirements will be documented in the applicable ADL (Authorized Data List). Such data will be delivered as identified on completed (numbered) DIDS when specified on DD Forms 1423 (Contract Data Requirements Lists) and incorporated into applicable contracts.

6.3 Supersession information. MIL-P-82650(OS) supersedes WS 14776 and is intended to be used in lieu of OS 10501 for Terrier rocket motor applications.

Source: https://assist.dla.mil -- Downloaded: 202610-23T07:19Z
Check the source to verify that this is the current version before use.

6.4 Preproduction sample waiver. Preproduction samples submitted and approved on a recent contract may be accepted by the procuring activity in lieu of an additional preproduction inspection. When the preproduction sample is waived (see 6.2.1(d)), the procurement document should contain a statement specifying that the standards of workmanship exhibited by the previously approved preproduction sample shall determine the minimum requirements of the current contract or order.

6.5 Suggested source of supply. A product which has met the requirements of this specification in past procurement actions is FM 16771-1 as manufactured by Fiberite Corporation, 501 W. Third Street, Winona, MN 55987. This information is for the convenience of the procuring activity and is not to be construed as a waiver of any requirement of this specification.

Custodian:
Navy - OS

Preparing Activity:
Navy - OS

Project No.
9330-N769

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:19Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/ vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE
MIL-P-82650(OS) "PLASTIC MOLDING MATERIAL, GLASS PHENOLIC"

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?
   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 68 WHICH MAY BE USED

S/N 0102-014-1802

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:19Z
Check the source to verify that this is the current version before use.

FOLD

Commanding Officer
Naval Ordnance Station (611)
Indian Head, MD 20640

POSTAGE AND FEES PAID



OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

Commanding Officer
Naval Ordnance Station (611)
Indian Head, MD 20640

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:19Z
Check the source to verify that this is the current version before use.